Assigment 8

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1) This problem will step you through making a prediction using more than one predictor variable

a) Modify Example 8.7 (continued) to predict the price of a car with 200 horsepower, a weight of 2705, and a length of 177. Use k = 3. You should get 19.533.

```
## [1] 19.53333
```

b.) Unfortunately this isn't what I obtained when I tested it with my homemade code. I wondered if this is because knnreg() isn't standardizing the predictor variables, so I tested it this way: I made a subset of the Cars93 dataset only containing the predictor variables, scaled it and turned it back into a data frame, then added the Price variable. I fit a model used this scaled dataset. Then I scaled the observation I wanted to predict for by the predictor variables and coerced it to a data fame, and finally made a prediction. It was 17.4. Obtain this same prediction.

```
## [1] 17.4
```

2.) (Grad student only) Write a function that implements Algorithm 8.6. Use it to solve problem 1) and ensure your answer is the same as in 1b).

Reproduce 1a with manual KNN (no standardization)

```
## [1] 19.53333
```

Reproduce 1b with manual KNN (with standardization)

```
## [1] 17.4
```